Local Choi population and qubit flip probability

A sharp constraint from trace preservation and its application to incomplete output data

Statement

Let C be the trace-one Choi matrix of a completely positive, trace-preserving two-qubit channel. Normalize vectorization by $|K\rangle = \mathrm{vec}(K)/2$. Let w be the Choi population in the subspace spanned by operators $V \otimes I_2$, and β the probability of measuring 1 on the second qubit following the input $|00\rangle$. Then

$$4(1-w) \geq \beta + \tfrac{1}{2}(1-\sqrt{(1-\beta)})^2.$$

For every β in [0,1] a channel attains equality. This is a sharp constraint on w and β considered by themselves.

Proof

Write each Kraus matrix in blocks A_j, B_j, C_j, D_j in row order, using the second bit for the block index and the first bit within each block. Orthogonal projection onto $V \otimes I_2$ replaces both diagonal blocks by $(A_j+D_j)/2$ and sets the other blocks to zero. With Frobenius norms it follows that

$$4(1-w) = \Sigma_j \left[\tfrac{1}{2}\|A_j-D_j\|^2 + \|B_j\|^2 + \|C_j\|^2\right].$$

Define $\alpha = \Sigma_j \|B_j|0\rangle\|^2$. Likewise $\beta = \Sigma_j \|C_j|0\rangle\|^2$. Trace preservation on $|00\rangle$ and $|01\rangle$ gives $\Sigma_j \|A_j|0\rangle\|^2 = 1-\beta$ and $\Sigma_j \|D_j|0\rangle\|^2 = 1-\alpha$. Restrict the Frobenius norms to their first columns, concatenate over j, and apply the reverse triangle inequality. The right-hand side is at least

$$\beta + \alpha + \tfrac{1}{2}(\sqrt{(1-\alpha)} - \sqrt{(1-\beta)})^2.$$

For fixed β this expression is increasing in α. On $0 \leq \alpha < 1$ its derivative is $\tfrac{1}{2} + \sqrt{(1-\beta)}/(2\sqrt{(1-\alpha)})$, and the endpoint follows by continuity. Substituting $\alpha = 0$ proves the inequality.

Attainment

In computational basis order take $K_0 = \mathrm{diag}(\sqrt{(1-\beta)},1,1,1)$ and $K_1 = \sqrt{\beta}\,|01\rangle\langle 00|$. They satisfy $\Sigma_j K_j^\dagger K_j = I_4$. The off-diagonal blocks contribute β and the diagonal-block difference contributes exactly $\tfrac{1}{2}(1-\sqrt{(1-\beta)})^2$.

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Consequence for the supplied model

Application to a calibrated rank-one effect and first-qubit tomography on the same channel

Local observations and purity

Let a be the larger eigenvalue of the first output marginal on $|00\rangle$, $b = \min(\beta,1-\beta)$, and $m = \min(1,a+b)$. When those local observations are the only data, a unitary is possible exactly for $1-a \le \beta \le a$. Necessity follows from the equal marginal spectra of a pure bipartite output. Sufficiency follows by rotating the second Schmidt basis to give β and extending the resulting output vector to a unitary.

The local-data upper bound is $H(m) = [9+6m+m^2+(1-m)^2]/16$. To see this, let Ω be the joint output. The positive block-matrix inequality bounds its maximum eigenvalue by the sum of the two diagonal-block norms, at most a+b. Thus (m,1−m,0,0) majorizes its spectrum. For a channel isometry, the environment states τ_j on the four basis inputs obey $16\,\mathrm{Tr}(C^2) = \mathrm{Tr}[(\Sigma_j \tau_j)^2]$. Here τ_0 has the spectrum of Ω up to zeros. Bounding the other three states by purity one gives $16\,\mathrm{Tr}(C^2) \le \mathrm{Tr}(\Omega^2)+6\lambda\max(\Omega)+9$.

Conditional numerical ceiling

In the supplied normalization, let $p = |I_4\rangle$, $u = |iZ \otimes I_2\rangle$, $v = |iX \otimes I_2\rangle$ and E project onto $\sqrt{h}\,p + \sqrt{(1-h)}u$. Write s, f for the p,v populations and $q = \mathrm{Tr}(EC)$. The reverse triangle inequality gives $w \ge s+f+(\sqrt{q}-\sqrt{(hs)})^2/(1-h)$. For $s=131/320$, $h=99/100$, $q=81711338569043/224995905216000$ and the specified 1% residual point, this yields $w \ge 0.99$ and the reference $a \approx 0.529472934504$.

Put $x = \sqrt{(1-\beta)}$. Page 1 gives $(x+1)^2 \ge 4-8(1-w)$, whence $\beta \le 1-(\sqrt{3.92}-1)^2 < 0.039797974644667$. H is increasing for the relevant m. Substitution using the exact inputs gives the outward-rounded ceiling

$$\mathrm{Tr}(C^2) \le 0.807826398288.$$

This improves the previous ceiling 0.807905661518. Attainment on page 1 fixes only the tradeoff between w and β. Its example does not establish equality under the separate detector, projector-population and first-marginal constraints. The sharp purity maximum for the full data is open. No matched device measurements or calibrated transfer across preparations are supplied.

Mathematical inputs: Purity Bounds from Local Output Measurements, pp. 1–3. Channel representation: J. Watrous, The Theory of Quantum Information, chapter 2. Pure marginal spectra: A. Higuchi, A. Sudbery and J. Szulc, Phys. Rev. Lett. 90, 107902 (2003).

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